

Aniket Mishra

(587)-937-3508 | ianiket23@gmail.com | linkedin.com/in/ianiket23 | github.com/ianiket23

EXPERIENCE

Software Engineer

June 2025 – Present

Total Life

Vancouver, BC (Remote)

- Developed an asynchronous data processing engine in **Go** utilizing **goroutines**, **buffered channels**, and **worker pools** to ingest and process high-volume record streams, improving throughput by **25%**.
- Architected a distributed **Pub/Sub messaging framework** to orchestrate automated data delivery across webhooks, external APIs, and notification systems, reducing manual handling by **31%**.
- Automated end-to-end data synchronization pipelines between cloud databases and CRM platforms, integrating **HubSpot** and **Stripe** webhooks to reduce monthly pipeline errors by **6%**.

Software Engineer - Contract

September 2024 – December 2024

vrCAVE

Edmonton, Alberta

- Optimized low-level hardware state management and multithreaded execution in **C++**, eliminating a 5-second frame stall and reducing average frame latency by **7%**.
- Designed custom state-synchronization routines across distributed hardware clients, **reducing multiplayer latency spikes by 20%** verified via Unreal Engine profiling.
- Refactored legacy codebases using strict design patterns and delegates to improve code maintainability, automated testing, and multi-client system stability.

Software Developer Intern

May 2022 – August 2022

G2V Optics

Edmonton, Alberta

- Developed a **solar tracking** feature for aerospace ground testing systems that computes mathematical models of **solar irradiance** based on coordinates (latitude, longitude, elevation) and time.
- Created a **bash script** to automate Raspberry Pi setup and **apache environment configuration** on Linux, reducing a 30-minute manual process to under **10 minutes**.

Software Developer Intern

January 2022 – April 2022

AI Shading

Calgary, Alberta

- Engineered real-time sensor data streaming via **WebSockets** for Tuya SDK and Open Weather APIs, reducing telemetry dashboard update latency from **250ms to 150ms**.
- Utilized **Datadog** for runtime monitoring, error profiling, and log pattern analysis to isolate and resolve recurring bugs, reducing runtime issues by **16%**.

SCIENTIFIC & SOFTWARE PROJECTS

NotExo.earth | *NASA Kepler Dataset, Python, Scikit-learn, Docker, GCP, React*

[\(Website\)](#)

- Founded a volunteer-led software initiative building automated data processing pipelines to analyze and visualize **open datasets from major space agencies** for exoplanet research.
- Developed interactive scientific visualizer to render Gaussian model distributions of solar wind interaction data from NASA's **MAVEN** orbiter and vSwim.
- Cross-calibrated multi instrument data from two different Mars orbiters (**MAVEN** and **Tianwen-1**), to minimize model uncertainty and improve the precision of MAVEN Gaussian fits.

FissionMunk | *Python, NumPy, pytest, PyPI, Poetry, GitHub Actions*

[\(PyPI\)](#)

- Developed and published an open-source **2D physics simulation library** on PyPI to model nuclear reactor mechanics and particle behavior.
- Implemented numerical integration algorithms with **NumPy** and integrated **WebSocket support** to stream simulation step states as JSON for real-time visualization.
- Enforced project-level development standards with **100% automated test coverage** using **pytest** and automated **GitHub Actions CI/CD pipelines**.

EDUCATION

University of Alberta

Edmonton, Alberta

Bachelor of Science with Specialization in Computer Science | GPA: 3.7

September 2019 – June 2024

- Dean's Honor Roll (2021–2024), International Student Scholarship

Alberta Machine Intelligence Institute (Amii)

Edmonton, Alberta

AI Career Accelerator Program

May 2026 – Present

TECHNICAL SKILLS

Languages & Core Systems: Python, C++, C, Go, Java, Bash/Shell, SQL, JavaScript, HTML/CSS

Scientific Computing & ML: NumPy, SciPy, Matplotlib, Scikit-learn, SPICE toolkit

Data Infrastructure & Web: Docker, Google Cloud Platform (Cloud Run, BigQuery), PostgreSQL, WebSockets, REST APIs, FastAPI, Next.js, React

Tooling & Practices: Git, CI/CD (GitHub Actions), Pytest, Datadog